

MSc Course in

Optimization Theory and Modern Reliability from a Practical Approach

Sample exam assignments (Reliability part), by
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Rules

- May give 2 to 4 tasks (can be solved without using special software tools, apart from Excel)
- The reliability part counts for 50% score of the course, with another 50% from the optimization theory part
- Time is flexible within the framed 4 hours of the course exam
- All usual helping aids are allowed, i.e. books, notes, calculator, computer, etc.
- All communication equipment and computer communication protocols must be turned off.
- Use of Internet is not allowed.
- Questions should be answered in English.

Tasks

Some errata have been highlighted

Task No. 1

A bronze bushing of a car gearbox is guaranteed for 3,000 operational cycles with normal $\Delta T = 90\text{ }^{\circ}\text{C}$. The number of cycle to failure versus ΔT follows an inverse power law with an exponent constant of -4.75. Find the equivalent end of life at severe operation condition of $\Delta T = 125\text{ }^{\circ}\text{C}$, assuming one cycle per day.

Task No. 2

An electronic power supply for laptop PC comprises:

- A. a main fuse
- B. a 4-diode main rectifier
- C. 3 dc-link capacitors in parallel
- D. 2 main power switches
- E. a transformer
- F. a 2-diode secondary rectifier
- G. an output filter
- H. a power indicator LED

Hypotheses:

1. All the above parts but the LED are needed to operate correctly.
2. Diodes 1 and 3 or diodes 2 and 4 are sufficient to make the main rectifier operate correctly.
3. 2 dc-link capacitors out of 3 are sufficient to keep the power supply working;
4. Both main switches are needed to properly operate;
5. The transformer and the output filter can be supposed to have infinite lifetime;

6. One diode in the secondary rectifier is sufficient to properly operate;
7. all the remaining parts (e.g. power cord, plastic enclosure, laptop cord & plug) are supposed to have infinite lifetime in respect to the ones listed above.

- 1) Sketch up the reliability block diagram of the system.
- 2) The failures of each subsystem are distributed according to the following statistical distribution:
 - A – Exponential, $\lambda = 20.0$ failures per million hours
 - B (single diode) – Weibull, $\beta = 1.5$, $\eta = 30.000$ hours
 - C (single capacitor) – Normal, $\mu = 20000$ hours, $\sigma = 3000$ hours
 - D (single switch) – Lognormal, $\mu = 9$, $\sigma = 4$
 - ~~E – Weibull, $\beta = 0.9$, $\eta = 10.000$ hours.~~
 - F (single diode) – exponential, $\lambda = 175$ failures per million hours
 - ~~G – Normal, $\mu = 20000$ hours, $\sigma = 5000$ hours~~
 - H – Exponential, $\lambda = 10.0$ failures per million hours

Calculate the system reliability at:

- a) $t = 1$ month;
- b) $t = 1$ year.

Task No. 3

For the system introduced in Task No. 2:

- 1) Try to identify cut sets and tie sets.
- 2) Starting from the found cut sets, try to build up an FTA (fault-tree analysis) of the event “PC power supply not working”

Task No. 4

The bolts of a horizontal heat sink experience cyclic stress at the base-plate connection due to the thermal expansion of the heat sink itself.

Accelerated data was taken on bolts where a cyclical stress was applied to the bolts whereby the metal at the base plate came under a continuous cyclical stress range of -6.0 MPa to +6.0 MPa with a mean tensile stress of zero. The bolts failed at the base plate connection after 10,000 cycles. Assuming a power-law exponent of $m = 3.5$ for the cycling, a metal tensile strength of 7.4 MPa and no defined elastic range:

- a) Estimate the number of cycles-to-failure CTF that would be expected in real use if the stress is between -2.5 MPa to +2.5 MPa.
- b) Estimate the number of cycles-to-failure CTF that would be expected in real use if the stress is between -2.5 MPa to +2.5 MPa but the heat sink is mounted vertically, producing mean tensile offset of $r_{\text{mean}} = 1.8$ MPa.

Task No. 5

A gear was tested in the laboratory to determine its fatigue life. The test results were as follows:

Stress level ($\times 10^2$ Nm)	3.4	7.2	12.0	15.3
Mean cycles to failure ($\times 10^6$)	10.9	4.5	0.76	0.23

The gear will operate in a real gearbox with the most relevant stress levels occurring in the following proportions, respectively:

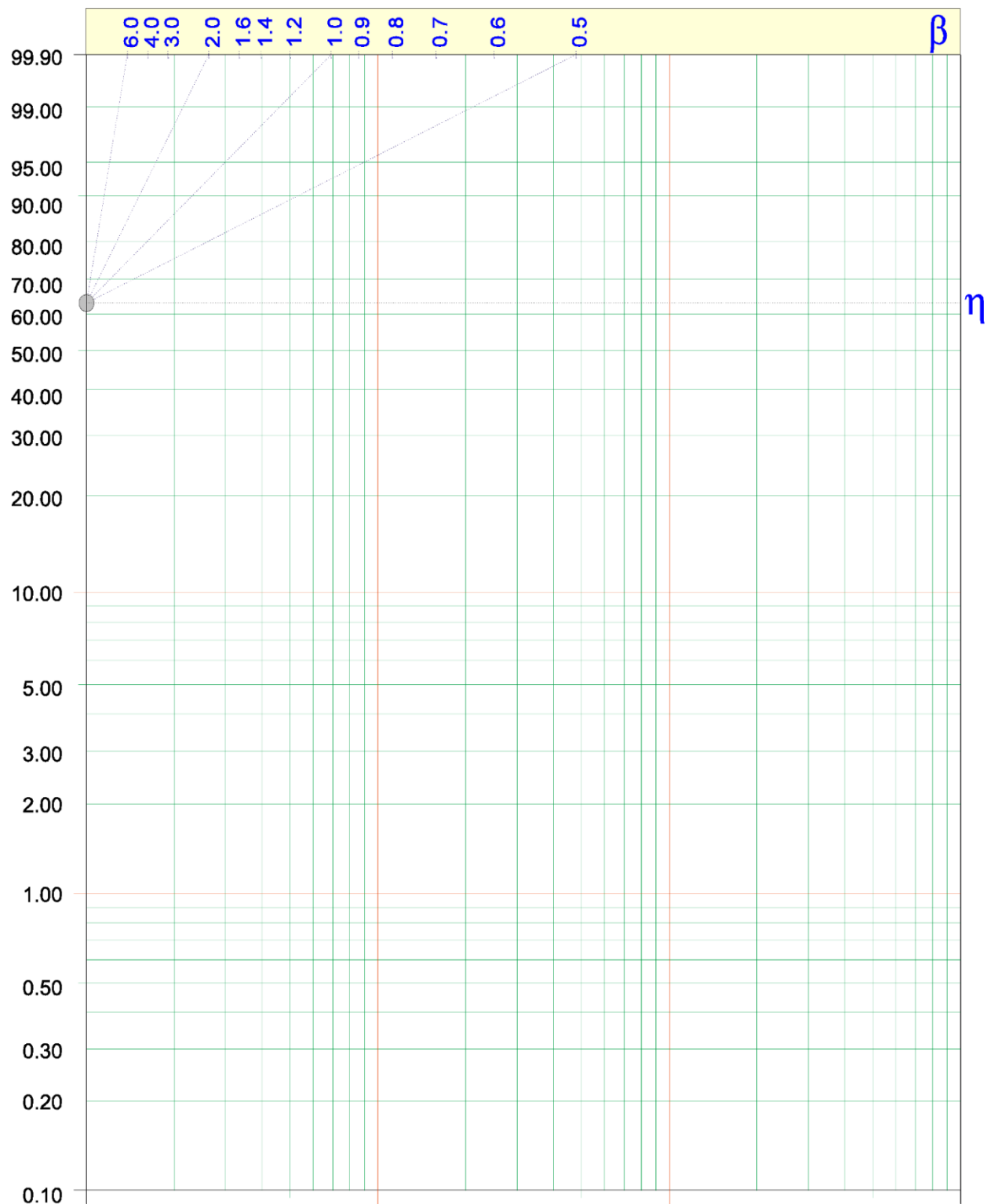
0.5 0.3 0.15 0.05

- ## Task No. 6

- 1) State the relationship between accumulated failure and time in Weibull distribution.
- 2) Arrange these time-to-failure numbers by using median ranking method.
- 3) Plot the ranking number as listed in (b) and the cycles to failure in the attached Weibull paper below. Please find out the values for β and η , respectively.
- 4) Explain how to identify the values of β and η in the Weibull plot.

failure rank = i

[illegible]



Task No. 7

Compare reliability values for the two products, Product A with exponentially distributed life and product B with Weibull distributed life. The parameters are $MTBF_A = \eta_B = 1000$ h. Compare the reliabilities at 300 hours for:

$$\beta_B = 0.5$$

$$\beta_B = 1.0$$

$$\beta_B = 3.0$$

How would you describe the effect of the Weibull shape parameter β on the reliability if the scale parameter remains the same?

Solutions to some of the above tasks

Task No. 1

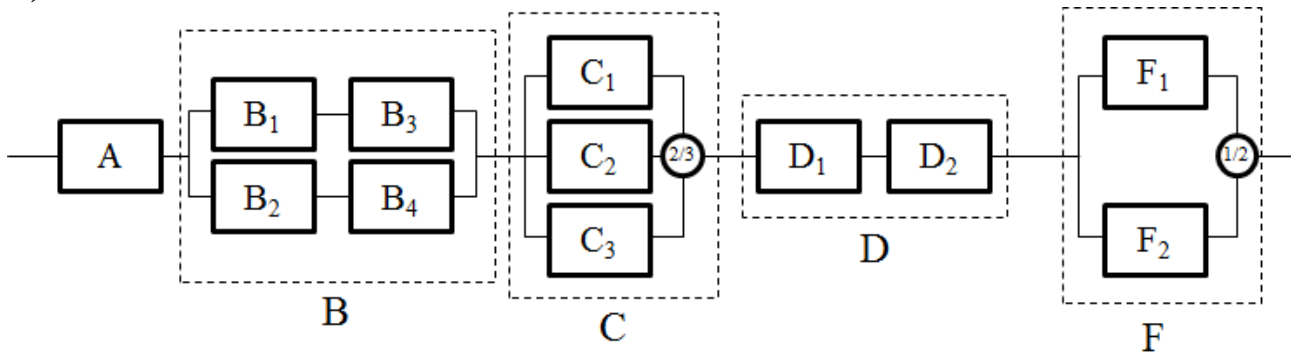
Define the normal life (in terms of number of cycles) as L_{normal} , and the life at the given severe condition is L_{severe} , the given exponent constant $m = 4.75$, $L_{normal} = 3,000$ cycles, the temperature variation in normal operation $\Delta T_{normal} = 90^\circ\text{C}$ and in severe condition $\Delta T_{severe} = 125^\circ\text{C}$. Therefore,

$$\frac{L_{severe}}{L_{normal}} = \left(\frac{\Delta T_{severe}}{\Delta T_{normal}} \right)^{-m}$$

$$L_{severe} = L_{normal} \times \left(\frac{\Delta T_{severe}}{\Delta T_{normal}} \right)^{-m} = 3000 \text{ cycles} \times \left(\frac{125^\circ\text{C}}{90^\circ\text{C}} \right)^{-4.75} \cong 630 \text{ cycles}$$

Task No. 2

1) The RBD can be drawn as follows:



Note that there is no reference to blocks E (transformer) and G (output filter) as they are supposed to have infinite life time, and H (LED) as its malfunction does not affect the overall system reliability.

2) For ease of calculation, it is convenient to work out the reliability of single blocks first, then compose with each other. Referring to Figure 2.11 of the book, “Shapes of common failure distributions, reliability and hazard rate functions” and using Microsoft Excel, one obtains for blocks A, B, C, D and F:

	A	B	C	D
1				
2			1 month	1 year
3	Block	Formula	720 hours	8760 hours
4	R_A	$\text{EXP}(-20/1000000 * D\$3)$	0,986	0,839
5	R_{B1-4}	$\text{EXP}(-((D\$3/30000)^{1,5}))$	0,996	0,854
6	R_{C1-3}	$1 - \text{NORMDIST}(D\$3; 20000; 3000; \text{TRUE})$	1,000	1,000
7	R_{D1-2}	$1 - \text{LOGNORMDIST}(D\$3; 9; 4)$	0,727	0,492
8	R_{F1-2}	$\text{EXP}(-175/1000000 * D\$3)$	0,882	0,216

Working out the reliability of complex blocks one obtains the following formulae:

a) For block B, it is the parallel of blocks in series:

$$R_B = 1 - (1 - R_{B13}) \cdot (1 - R_{B24}) = 1 - (1 - R_{B1}R_{B3}) \cdot (1 - R_{B2}R_{B4}) = 1 - (1 - R_{B1-4}^2)^2$$

b) For block C, it is the redundancy m-out-of-n:

$$R_C = 1 - \sum_{i=0}^{2-1} \binom{3}{i} R_{C13}^i (1 - R_{C1-3})^{3-i} = 1 - [(1 - R_{C1-3})^3 + 3 \cdot R_{C1-3}(1 - R_{C1-3})^2]$$

c) For block D, it is the series of two blocks:

$$R_D = R_{D1}R_{D2}$$

d) For block F, it is the parallel of 2 blocks:

$$R_F = 1 - (1 - R_{F1-2})^2$$

Using Excel once again, one obtains:

	1 month	1 year
	720 hours	8760 hours
R_A	0,99	0,84
R_B	1,00	0,93
R_C	1,00	1,00
R_D	0,53	0,24
R_F	0,99	0,39
$R_A \times R_B \times R_C \times R_D \times R_E \times R_F$	0,51	0,07

In conclusion, one obtains:

a) $R(t = 1 \text{ month}) = 51\%$

b) $R(t = 1 \text{ year}) = 7\%$

Task No. 4

From the given data [$\Delta\sigma = 6.0\text{MPa} - (-6.0 \text{ MPa}) = 12.0 \text{ MPa}$; $\Delta\sigma_{\text{elastic}} = 0$; $m = 3.5$; $\text{CTF} = 10.000$] we can calculate B_0 rearranging (1) in (2):

$$\text{CTF} = B_0(\Delta\sigma - \Delta\sigma_{\text{elastic}})^{-m} \quad (1)$$

$$B_0 = \frac{\text{CTF}}{(\Delta\sigma - \Delta\sigma_{\text{elastic}})^{-m}} = \frac{10.000}{(12.0 - 0)^{-3.5}} = 5.99 \cdot 10^7 \quad (2)$$

a)

$$\text{CTF} = 5.99 \cdot 10^7 \cdot (2.5 \text{ MPa} - (-2.5 \text{ MPa}))^{-3.5} \cong 210\,000 \text{ cycles}$$

b)

From theory in case of stress bias:

$$\text{CTF} = B_0 \left[\frac{\Delta\sigma}{1 - (\sigma_{\text{mean}}/\sigma_{\text{TS}})} - \Delta\sigma_{\text{elastic}} \right]^{-m} \quad (3)$$

Using (3) one can obtain:

$$CTF = 5.99 \cdot 10^7 \cdot \left[\frac{2.5 \text{ MPa} - (-2.5 \text{ MPa})}{1 - \frac{1.8 \text{ MPa}}{7.4 \text{ MPa}}} - 0 \right]^{-3.5} \cong 81.000 \text{ cycles}$$

Task No. 5

The expression of Miner's rule for a reference period (e.g. 1 year) is the following:

$$TCL = \sum_{i=1}^k \frac{n_i}{N_i} \quad (1)$$

Where TCL is the "total consumed lifetime", k is the number of different stress levels, N_i is the number of cycles to failure at i-th stress level and n_i is the number of cycles performed in reality at i-th stress level.

For a reference period, e.g. 1000 cycles, using Miner's rule it can be easily concluded:

$$TCL|_{1000 \text{ cycles}} = \frac{0.5 \cdot 1000}{10.9 \cdot 10^6} + \frac{0.3 \cdot 1000}{4.5 \cdot 10^6} + \frac{0.15 \cdot 1000}{0.76 \cdot 10^6} + \frac{0.05 \cdot 1000}{0.23 \cdot 10^6} = (4,58 + 6,66 + 19,7 + 21,7) \cdot 10^{-5} = 0,000526 = 0,0526\% \quad (2)$$

The number of cycles to failure is the number of cycles to get a 100% consumed lifetime, hence:

$$CTF = \frac{100\%}{0,0526\%|_{1000 \text{ cycles}}} = 1899 \times 1000 \text{ cycles} = 1,899 \cdot 10^6 \text{ cycles} \quad (3)$$

a) In case of 100 cycles per hour, one obtains:

$$TTF = \frac{1,899 \cdot 10^6 \text{ cycles}}{100 \text{ cycles/hour}} = 18990 \text{ hours} (\cong 2.1 \text{ years})$$

b) In case of 1000 cycles per hour, one obtains:

$$TTF = \frac{1,899 \cdot 10^6 \text{ cycles}}{1000 \text{ cycles/hour}} = 1899 \text{ hours} (\cong 79 \text{ days})$$

Task No. 6

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^\beta\right]$$

1)

2) See Table 1 in the attached Weibull paper on next page.

3) $\beta=1.6$ (1.2-3); $\eta=50$ (20-100). These ranges can be acceptable due to the error of the curve fitting.

4) β is obtained by using parallel translation of the fit line to the β -nomogram in the Weibull plot. As shown in Weibull expression, regardless of the variation of β , $F(t)$ becomes 63.2% if t equals η . That is why η refers to the cycles when failure function reaches 63.2%.

Task No. 7

Exponential

$$R(300) = \exp\left(-\frac{300}{1000}\right) = 0.7408$$

Weibull:

a. $\beta = 0.5$

$$R(300) = \exp\left[-\left(\frac{t}{\eta}\right)^\beta\right] = \exp\left[-\left(\frac{300}{1000}\right)^{0.5}\right] = 0.5783$$

b. $\beta = 1.0$

$$R(300) = \exp \left[- \left(\frac{300}{1000} \right)^{1.0} \right] = 0.7408$$

c. $\beta = 3.0$

$$R(300) = \exp \left[- \left(\frac{300}{1000} \right)^{3.0} \right] = 0.9734$$

The higher the Weibull slope the higher the reliability at time before the characteristic life ($t < \eta$). The opposite is true if $t > \eta$.

